

Chapter 6: 4th and 5th editions
Section 6.1, pp. 161-163, Nos. 5, 6, 7

Due Date: 04/24/2013 12:00 PM
Section 6.2, p. 169, Nos. 4

5) Eliminate all useless productions from the grammar

$S \rightarrow aS|AB,$
 $A \rightarrow bA,$
 $B \rightarrow AA.$

Step 1: Remove Nullable Variables
No lambda productions.

Step 2: Remove Unit-Productions
No unit productions.

Step 3: Remove Useless Variables
 $A \rightarrow bA$ does not generate any string, and therefore.
 B does not generate any string either since $B \rightarrow AA$.
 $S \rightarrow aS$ does not generate any string either.

What language does this grammar generate?

Since none of the variables in the grammar generate a string then the grammar is useless because $S \rightarrow aS$ and $A \rightarrow bA$ always loop. If there was λ at the end of each production the language would have looked something like this $a^* U b^*$.

6) Eliminate useless productions from

$S \rightarrow a|aA|B|C,$
 $A \rightarrow aB|\lambda,$
 $B \rightarrow Aa,$
 $C \rightarrow cCD,$
 $D \rightarrow ddd$

Step 1: Remove Nullable Variables

$S \rightarrow a|aA|B|C,$
 $A \rightarrow aB|a,$
 $B \rightarrow Aa|a,$
 $C \rightarrow cCD,$
 $D \rightarrow ddd$

Step 2: Remove Unit-Productions.

$S \rightarrow a|aA|Aa,$
 $A \rightarrow aB|a,$
 $B \rightarrow Aa|a,$
 $D \rightarrow ddd$

Step 3: Remove the useless variables.

$S \rightarrow a|aA|Aa,$
 $A \rightarrow aB|a,$
 $B \rightarrow Aa|a$

Result:

$S \rightarrow a|aA|Aa,$
 $A \rightarrow aAa|aa|a$

7) Eliminate all useless productions from

S $\rightarrow$ **AaB|aaB**

A $\rightarrow$ λ ,

B $\rightarrow$ **bbA| λ .**

Step 1: Remove Nullable Variables

S $\rightarrow$ **aB|a|aaB|aa**

B $\rightarrow$ **bb.**

Step 2: Remove Unit-Productions.

No unit production.

Step 3: Remove the useless variables.

S $\rightarrow$ **aB|aaB|aa|a**

B $\rightarrow$ **bb.**

4) Transform the grammar with productions

S $\rightarrow$ **abAB,**

A $\rightarrow$ **bAB| λ ,**

B $\rightarrow$ **BAA|A| λ**

First: Removing λ productions:

S $\rightarrow$ **abAB|abA|abB|ab**

A $\rightarrow$ **bAB|bA|bB|b**

B $\rightarrow$ **BAA|Ba|Aa|a|bAB|bA|bB|b**

Second: Introduce variables C and D to substitute terminals a and b.

S $\rightarrow$ **CDAB|CDA|CDB|CD**

A $\rightarrow$ **DAB|DA|DB|b**

B $\rightarrow$ **BAC|BC|AC|DAB|DA|DB|b|a**

C $\rightarrow$ **a**

D $\rightarrow$ **b**

Third: Introduce variables to shorten the right sides of the production.

S $\rightarrow$ **EF|EA|EB|CD**

A $\rightarrow$ **DF|DA|DB|b**

B $\rightarrow$ **BG|BC|AC|DF|DA|DB|b|a**

E $\rightarrow$ **CD**

F $\rightarrow$ **AB**

G $\rightarrow$ **AC**

C $\rightarrow$ **a**

D $\rightarrow$ **b**